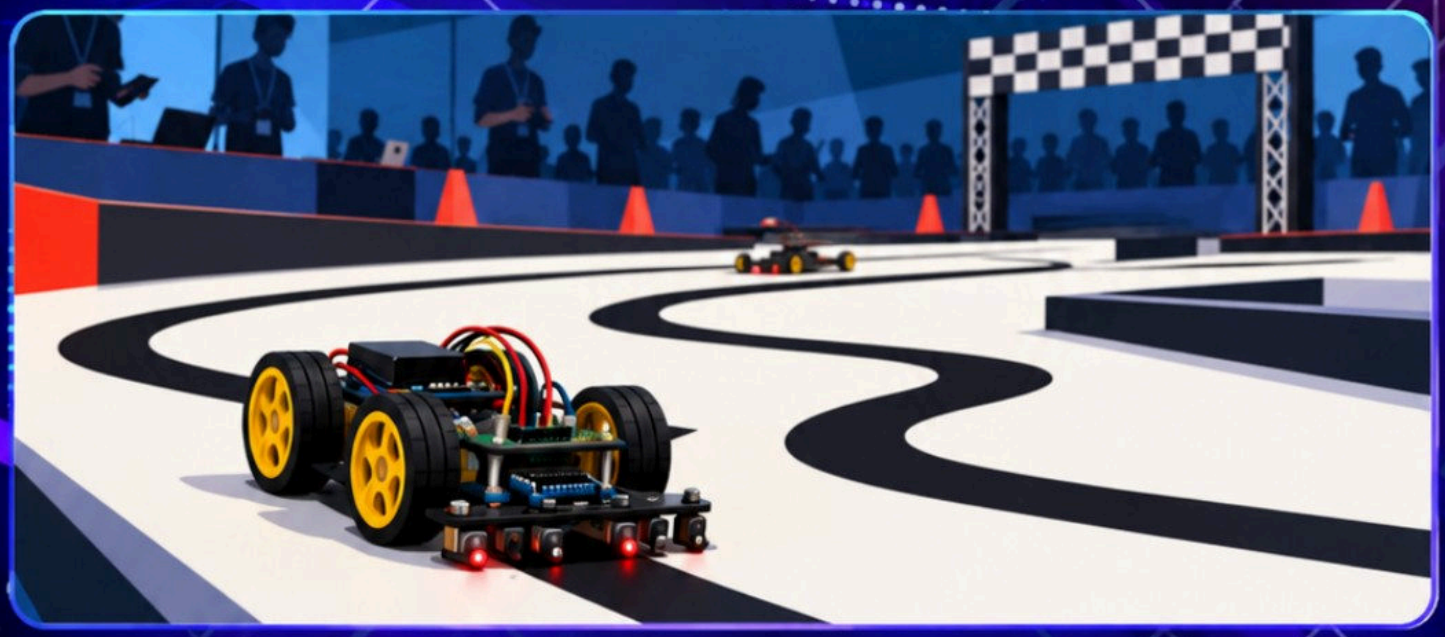
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LINE FOLLOWER



Introduction

Line Follower is a two-round Robotics & Embedded Systems competition designed to test participants' circuit-building skills, sensor calibration, control logic and ability to make a robot autonomously track a path within a fixed set of hardware specifications.
Teams of three members each will design, assemble and fine-tune a line-following robot built strictly from the components provided, ensuring the contest is a fair test of skill rather than hardware budget.

Event Description

Participants must build and tune an autonomous line-following robot capable of completing the prescribed tracks. The robot must operate using only the components specified and provided by the ECE Society. The competition emphasizes reliable sensing, motor control, circuit assembly and autonomous path tracking.

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Event Format

ROUND 1 — QUALIFYING ROUND

- Each team will consist of 3 members.
- A simple track will be used for this qualifying round.
- Teams only need to complete the track according to the given guidelines.
- There is no time limit in this round; successful completion of the track is the only qualifying criterion.

ROUND 2 — BEST OF TWO (COMPETITIVE ROUND)

- Qualified teams will compete on a more challenging track.
- Each team is given two chances (attempts) to complete the track.
- The better of the two timings will be considered for final ranking.
- If the robot loses the line at any point, it must restart from the last checkpoint rather than the beginning.

TRACK AND HARDWARE SPECIFICATION

TRACK SPECIFICATION

The event will use two printed line-follower tracks. The qualifying track will be relatively simple and will be used to assess the basic performance and reliability of the robots. The competitive track will be more challenging and will be used for timed ranking of the qualified participants.

- Track Surface: White background
- Line Color: Black
- Line Width: 20 mm
- Track Type: Printed line-following track

HARDWARE SPECIFICATION RULE

- All robots must be built using only the components listed and provided by the ECE Society.
- No extra or more powerful components may be used beyond the given specification.
- The restriction applies to motors, sensors, controllers, batteries and other relevant hardware.
- The purpose of the rule is to ensure a level playing field and make the contest a fair test of design skill rather than budget.

ALLOWED COMPONENTS (PROVIDED BY ECE SOCIETY)

- Arduino Uno
- IR Sensor Array
- Lithium-ion Cell (3.7V, 2200mAh)
- TP4056 Li-ion Charging Module
- Jumper Wires
- BO Motors
- L298N Motor Driver
- BO Motor Wheels
- Breadboard
- Chassis (Single)

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FAIR-PLAY AND HARDWARE RULES

1. Only the components specified and provided by the ECE Society may be used.
2. No additional or more powerful motors, sensors, controllers, batteries or other components may be introduced.
3. Teams must operate within the supplied hardware specification throughout the competition.
4. Participants are not allowed to touch, move, or manually guide the robot once the run has started.
5. The decision of the judges/organizing committee regarding timing, qualification, penalties and disqualification will be final and binding.
6. Any form of cheating, intentional interference, or violation of the competition rules may result in immediate disqualification.
7. In case of a technical problem with the official track or timing system, the judges may provide a re-run. Technical issues caused by the participant's own robot will generally not qualify for a re-run.
8. The organizers reserve the right to modify or clarify the rules if required for the smooth and fair conduct of the competition. Any such changes will be announced to all participants before implementation.

FINAL RANKING

Round 1 determines qualification through successful completion.
Round 2 determines the final ranking using the best timing from the two competitive attempts.

Prize Pool : 20K (Including goodies)

CONTACTS

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scan to join the watsaap group